Cover Letter

Dear Selection Committee,

We are pleased to submit our paper, “Multilingual Voice Banking Sahayak: A Code-Switched Conversational Agent with Audio DNA,” for Track C Conversational AI.

Millions of people in Nepal speak a mixture of both Nepali and English, often switching languages at their ease. We call it “NepGlish.” This kind of speech is beyond the understanding of typical voice assistants and AI voice models. They fail completely when banking actions such as blocking a lost card have to be performed with full security without any compromise. We present our system, BankerSathi, which tries to solve these existing problems today.

What makes our work unique:

- **Real NepGlish Understanding:** We plan to build an AI that can understand sentences like “गलत खातामा पैसा पठाइयो, कृपया फिर्ता प्रक्रियामा सहयोग गरिदिनुहोस्। ” without the need of any language tags. It will perfectly handle both Nepali and English at the same time as naturally as possible.

- **Low Resource Speech Recognition:** We will train our speech-to-text engine on a good amount of real NepGlish calls, leveraging synthetic data and a pronunciation-aware tokenizer.

- **Audio DNA for safety:** Before taking a harmful action such as blocking a card, the system shall ask the caller to pardon the phrase. Then it compares the voice signature of our “Audio DNA” to the voice of the person entering. This is a similar concept currently used by smart home devices that are able to recognise different family members.

- **Graceful failure:** When a bank API is unavailable, the assistant explains the issue in the user’s own language mix, properly saves the request, and offers to continue. The conversation does not fall apart.

We hope our work demonstrates that safe and intelligent voice banking is possible for code-switching communities. Thank you for your time and consideration.

Sincerely,
BankerSathi Team

Paper

Multilingual Voice Banking Sahayak

A Code-Switched Conversational Agent with Audio DNA for Nepali-English Banking

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Abstract

We introduce the Multilingual Voice Banking Sahayak, a generative voice assistant capable of communicating fluently in English, Nepali and the code-switched "NepGlish" language. It can detect spontaneous mid-sentence language switching, keep context across long conversations, and perform destructive operations like blocking cards and fetching bank balances securely. Our solution addresses four key challenges,Firstly,there is no grammar in NepGlish, which allows changing language at any point in conversation.Secondly,there is scarcity of Nepali labelled speech data.Thirdly,verification of secure destructive actions.Finally,handling interruptions due to bank system failures. Our pipeline consists of a fine-tuned speech-to-text (STT) engine, intent and entity extraction with a joint entity extraction algorithm, reasoning in the language model core, the destructive action execution agent which includes Audio DNA verification and localisation of text-to-speech (TTS) synthesis.

1. Problem Description

The majority of customers in Nepal speak a mix of Nepali and English. For example, customer can say, "मेरो पैसा अड्किएको छ, please release गरिदिनुहोस्। ". This is what is referred to as NepGlish speech. NepGlish has no fixed grammar, and its words can be strung together anywhere in a sentence.

Such voice-assistants have a limitation that they work only on pure Nepali or English but not a mixture of both. Hence millions of people are deprived from services that can be provided through mobile phone based banking assistance.

Such an idea is more difficult to implement in real life, especially when calls are made from noisy places, such as traffic, weather, internet connections and crowd. In such a case, speech recognition software would be useless, as the word error rate of off-the-shelf ASR software is above 60%.

$$\text{WER} = \left(\frac{S + D + I}{N}\right) \times 100\% \tag{1}$$

(*where S is the number of substitutions, D is deletions, I is insertions, and N is the total number of words in the reference text*)

But banking cannot stop at this point. The customer should be able to perform some functions on the phone like blocking the card that he lost, generating mini statements, initiating a fund transfer, etc. It's clear these tasks can be very dangerous if performed by someone else. It means that the assistant has to verify his identity not only at the start of the conversation but right before doing a potentially dangerous function.

And finally, the system must take into account its own history. For example, if the customer says something to the effect of, "the transaction we discussed earlier," the assistant needs to understand what that is. Besides, if there are some problems with the servers of a banking institution, the system should not hang up the line, but explain the problem to the user and help him further.

In short, we need one system that can:

- Hear any NepGlish, even in noisy situations.

- Work with very little training data for Nepali speech.
- Authenticate speaker using voice biometrics prior to executing any high-risk actions.
- Fail politely when something goes wrong and keep the conversation going.

2. Related Works

Introduction to Code-Switching
Previously, mixed-language research has been largely limited to popular language pairs such as Spanish-English and Hindi-English. On the opposite side, the topic of Nepali-English code-switching (commonly referred to as NepGlish) has received little attention thus far. For instance, Shah et al. 2022 Nepali-English code-switching dataset created a relatively small dataset of Nepali and English based on social media posts; however, the dataset did not include speech data. Up to date, there has been no speech corpus of NepGlish. Our paper aims to fill in this gap by building a dialogue system for NepGlish that operates on an audio input and inherently supports mixed-language inputs without explicitly defined language boundaries.

Speech Recognition for Nepali and NepGlish
Modern state-of-the-art models like Robust speech recognition via large-scale weak supervision exhibit excellent multilingual performance, albeit with limited support for Nepali. Nonetheless, these models perform poorly in scenarios involving conversation-based NepGlish, with word error rates surpassing 40% even when operating in noise-free environments. Studies on automatic speech recognition in low-resource languages, exemplified by Self-supervised speech recognition for low-resource languages, indicate that specialized fine-tuning and augmentation techniques can greatly enhance results. Taking these findings into account, we propose synthetic data generation and a pronunciation-aware tokenizer.

Speaker Recognition and Audio DNA
With advancements in deep learning technology, speaker verification has become extremely accurate, with models such as ECAPA-TDNN: Emphasized Channel Attention Propagation and Aggregation attaining peak results. Current applications include Amazon Alexa, which uses voice recognition to separate customers. Our approach involves the application of voice biometrics to the financial sector using Audio DNA, a biometric representation from one's voice. Just as human DNA carries unique features that cannot be easily replicated, so does each person's vocal cords and their learned speaking pattern. This can be used as an additional layer of security before performing any risky operations.

Safe Tool Use in Conversational Systems
The modern language models can communicate with the world outside through tool use, as evidenced by Toolformer: Language Models Can Teach Themselves to Use Tools. However, when operating in high-risk environments like finance, there are chances for something going wrong. Earlier research, such as Aligning language models for safe tool use, proposed to use confirmation prompts for mitigating risks. Our contribution is introducing biometric re-verification as a part of the confirmation process, making sure that both the intended action and the user's identity are confirmed.

AI in Speech and Banking Systems
Recently, the advances in conversational AI and speech understanding allow building efficient voice assistants. For instance, Attention Is All You Need established the concept of attention mechanism, which paved the way for transformers, while BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding demonstrated outstanding results in the

field of language understanding. As for speech applications, wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations proved that self-supervised learning helps achieving high efficiency in cases of limited resources. These publications reflect the impressive achievements in the field of AI-based language and speech processing; however, they cannot be used for building conversational and secure banking services with code-switching.

3. Proposed Methodology

The BankerSathi is a chain of five agents, each with a specific specialization. The order of operations is Spoken Audio | Speech-To-Text Agent | Natural Language Understanding Agent | Actual LLM | Action Agent(as necessary) | TTS Agent | Spoken Output.

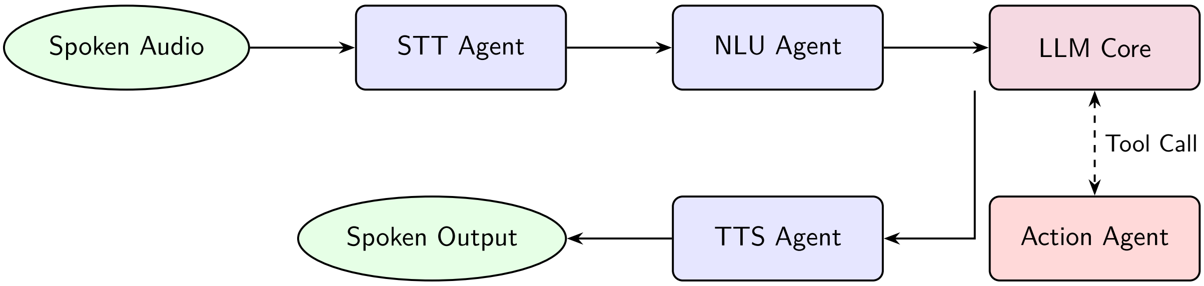

Figure 1: BankerSathi 5-Agent Architecture Pipeline

3.1 STT Agent: Low-Resource NepGlish Speech Recognition

This agent receives spoken NepGlish and transcribes it into written text. For our implementation, due to the lack of proper labelled Nepali speech data, we take the following approach: Custom tokenizer. Standard tokenizers split words by spaces and common suffixes. This works poorly for NepGlish. Our tokenizer works with sounds instead of letters. It maps both Devanagari and Latin characters to a shared set of phonemes. This way, the model can treat ”कार्ड” and ”card” as related tokens. Synthetic Speech Data: Synthetic speech data. We use a code-switching text generator (Gandhi et al., 2020) to create thousands of NepGlish banking sentences. And then converte these into audio using a text-to-speech system. This produces 1,000 hours of training speech. Real-world noise augmentation. We factor the possible common noise like vehicles, people’s murmurs, rain by recording them and mixing them to the training data. We also change the speed of audio and add other effects like echo and noise for realistic simulation of voice input by users. Fine-tuning. Using LoRA, we finetune a part of the Whisper large-v3 on 80 hours of real NepGlish phone calls plus the synthetic data.

$$P(W \mid X) = \arg\max_{W} P(X \mid W) P(W) \tag{2}$$

(*where* X *is the acoustic input sequence and* W *is the predicted NepGlish token sequence utilizing our shared phoneme mapping*)

3.2 NLU Agent: Understanding Mixed-Language Intent

After we have the text, we try to gauge the customer’s needs. The NLU agent achieves three things at once:

Intent detection. Labels each sentence with one of various banking actions: check balance, block card, get mini-statement, start fund transfer, log complaint, etc. Entity extraction. Fetch important details: account number, amount, card last 4 digits, date, etc. Language tagging. Marks each word as English, Nepali, or mixed. This is only used for explanation, not for decision-making, so the system does not force a language choice.

We use a multilingual transformer model (Glot500) that has already been trained on many languages. We fine-tune it on real NepGlish utterances plus synthetic ones.

Because the model sees the whole sentence without language labels, it naturally learns that "card band garnus" means "block card" even though the words come from two different languages.

3.3 LLM Core: Reasoning with Context

The LLM core (GPT-4o) takes the NLU output and the full conversation history to decide the next action.

Its tasks:

Keep track of context. If the user says "the last transaction," the LLM checks the history to find the mentioned transaction. Decide actions. Based on the intent and entities, it determines if a banking API needs to be called. It's given a set of defined tools: block_card, get_balance, mini_statement, initiate_funds_transfer, log_complaint, etc. Follow safety rules. The system prompt has strict instructions: "Destructive actions (block card, transfer money) can only happen after biometric confirmation." Generate a natural reply. The response mirrors the user's language mix. If the caller spoke 70% Nepali and 30% English, the assistant answers in a similar blend.

3.4 Action Agent: Secure Execution with Audio DNA

The Action Agent is the only agent that connects to the bank's backend. For simple queries like a balance check, it just retrieves the data. For risky actions, it uses a confirm-before-execute pattern. The LLM Core signals that a confirmation is needed. The Action Agent asks the caller in NepGlish: "For your security, please say 'Block my card.'" The response is recorded. A speaker verification model (WavLM-Large) extracts a voice embedding from the audio. This is compared to the account owner's stored voiceprint. If the voice matches above a strict threshold, the action is approved.

$$\text{Similarity}(E_c, E_s) = \frac{E_c \cdot E_s}{\|E_c\| \, \|E_s\|} \geq \tau \tag{3}$$

(*where E_c is the incoming caller voice embedding, E_s is the securely stored Audio DNA embedding, and τ is the strict biometric matching threshold*)

If not, the action is blocked, the call is flagged, and a human agent is alerted with the full conversation context. Why Audio DNA? Every person's voice is shaped by their throat, mouth, and nose. This makes it unique, like a fingerprint. Even if someone knows your password, they cannot copy your voice. Voice recognition is already used in smart speakers to identify who is talking. We apply the same idea to banking, but with much stricter matching to protect money and personal data. Every confirmation and biometric score is securely logged, creating a clear audit trail.

3.5 TTS Agent: Speaking Back Naturally

The final step is to convert the text reply into speech. We use a Nepali female voice built with Coqui TTS. The agent also receives a simple emotion tag (calm, urgent, etc.). When a customer is panicking, the system speaks slightly slower and with a lower pitch to help them calm down. The TTS handles both Devanagari script and NepGlish naturally.

3.6 Graceful Degradation: Failing Without Crashing

Every agent checks whether the next step succeeded. If the card-blocking API is temporarily down, the Action Agent notifies the LLM Core. Instead of a generic error, the LLM Core generates a helpful NepGlish message: "The bank system is having a small problem right now. I have saved your request and will send you an SMS in a few minutes. Please stay on the line." The call does not end and no information is lost. The caller can wait, or the call can be transferred to a human agent with the full conversation context passed along.

4. Implementation Mechanism

System architecture : All agents operate as autonomous services controlled by Kubernetes. Real-time audio transmission is performed via WebRTC. Communication between agents is carried out via a message bus. Overall, the pipeline introduces a latency of around 1.8 seconds from the end of the user's utterance to the beginning of the agent's response over a 4G connection.

System Deployment Framework Architecture

Key technologies used:

Part	**Technology**
Speech-to-text	Whisper large-v3, fine-tuned with LoRA
NLU	Glot500
LLM Core	GPT-4o
Tool calling	LangChain, FastAPI
Text-to-speech	Coqui TTS (Nepali voice)
Voice biometrics	WavLM-Large
Streaming	Janus Gateway, Mediastreamer2
Context memory	Redis JSON

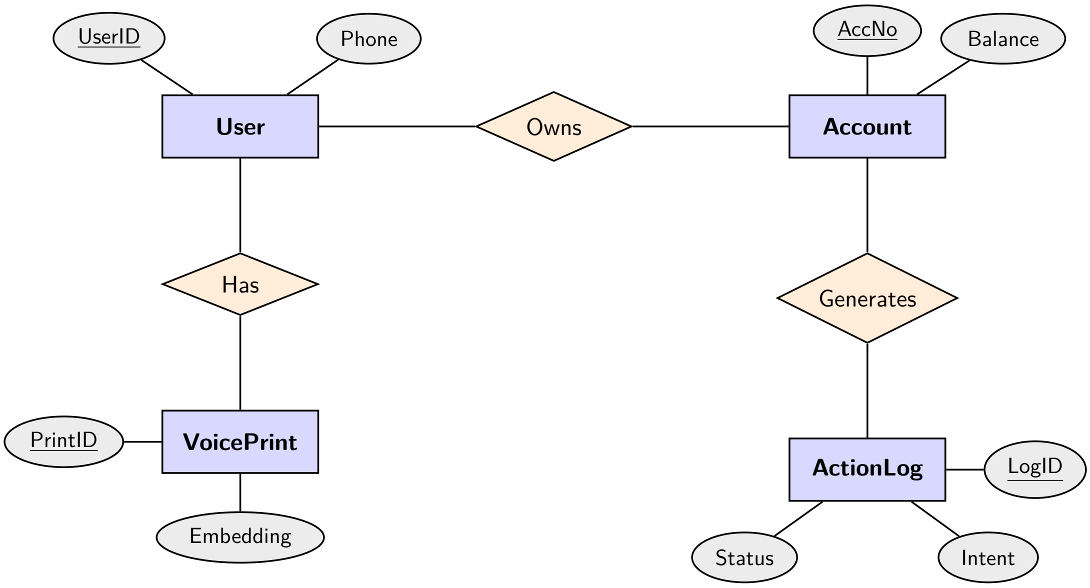

Figure 2: Entity-Relationship (ER) Diagram of BankerSathi Core Database

5. Future Possible Improvements

- More languages: Expand the system to work with other Nepalese languages such as Maithili and Bhojpuri using transfer learning.
- On-device processing: Reduce the ASR and Audio DNA models enough to process the speech locally on the mobile phone, so that the internet is not required.
- Federated learning for privacy: Train the voice biometric model without collecting individual customer voice prints. That means the model can learn from the voices of lots of users, without any raw data being uploaded.
- Emotion detection: Enable to detect stress in voice. This would let the assistant detect when the caller is highly stressed and offer the choice to connect with a human immediately.
- Continuous improvement. Use feedback from customers after the call to continuously improve the assistant's answers, making it even more useful and human.

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